LLM Hallucination Detector

Detecting Hallucinations via Attention Spectrum, Semantic Entropy, and Embedding Geometry

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The Problem

LLMs Fabricate Facts

Generative AI draws from probability distributions. Randomness can produce plausible-sounding but false statements, especially for rare or ambiguous prompts.

Costly to Detect

Current detection requires expensive LLM judges or human review. Companies deploying millions of queries need something much cheaper.

Agentic AI Amplifies Risk

Agentic systems make recurrent LLM calls across platforms. A single hallucination can cascade through multi-step reasoning chains.

Our Proposal: Two Approaches

Build a lightweight model that cheaply flags high-risk responses, so only flagged questions get elevated to expensive judges.

Part I: Attention Head Spectrum

Extract top eigenvalues from attention matrices at each layer. PCA compresses ~7,000 spectral features into a trainable representation. Requires model internals.

Part II: Response Cloud Geometry

Sample N=20 responses per question, embed with sentence transformers, compute 6 unsupervised geometric features. Works as a black box: no model internals needed.

Together they cover both white-box (internal access as Llama-3.2-3B) and black-box (API only) deployment.

Data Collection Pipeline

1 → 2 → 3 → 4

QA Benchmarks

5 datasets

500 questions each

DefAn, HaluEval, MMLU, TriviaQA, TruthfulQA

Llama-3.2-3B

N=1 (for Attentions)

N=20 responses per question (Semantic)

T=1.0, nucleus sampling

H100 GPU (Colab)

GPT-4.1-nano LLM Judge

Correct / Incorrect / Refused

Per-response labels

Majority vote for q-level

Probability “y” of hallucination

Collect Eigenvalues Embeddings + Features

Extract Attention Heads’ Laplacian Spectrum

all-MiniLM-L6-v2

6 geometric features

One row per question

Final dataset: 2,500 questions × 6 features. Ratio: 417 samples per feature.

Benchmark Datasets

Dataset	Questions	Difficulty	Notes
DefAn	500	Hard	Definitional QA, stable domains, has paraphrases
HaluEval	500	Easy	Curated pairs with context provided in prompt
MMLU	500	Hard	57-subject benchmark, verbose responses
TriviaQA	500	Medium	Trivia reading comprehension, balanced classes
TruthfulQA	500	Hard	Adversarial misconception questions, some refusals

Key EDA findings: HaluEval was the easiest (context in prompt). MMLU was hardest (>50% threshold rarely met). Distribution of correctness is bimodal across all datasets: the LLM either fails entirely or is entirely correct per each response cloud.

Refusals → label 1: Refusals were sparse and did not represent a substantially different failure mode. They are merged with incorrect for the binary hallucination label.

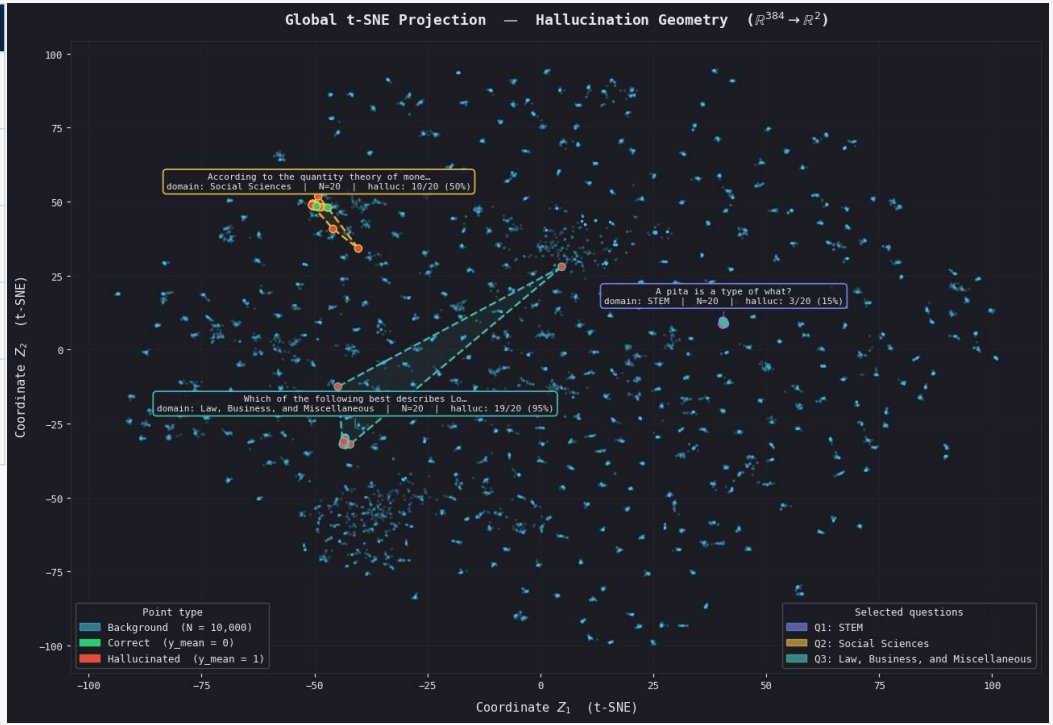
How It Works: Sampling a Single Question

Question from DefAn: "What is the definition of ontology?"

#	LLM Response (Llama-3.2-3B, T=1.0)	Judge
1	Ontology is the branch of philosophy concerned with the nature of being and existence.	Correct
2	Ontology is the study of what exists, dealing with questions of being and reality.	Correct
3	Ontology refers to the classification of living organisms in biology.	Incorrect
4	Ontology is a branch of metaphysics exploring the nature of existence.	Correct
. . .	... (16 more responses sampled at temperature 1.0) ...	...

From 20 responses: if 16 are correct and 4 are not, the hallucination rate is 4/20 = 0.20. Since 0.20 < 0.50, the question-level label is y = 0 (correct). The 6 geometric features are then computed from all 20 response embeddings.

T-SNE projecting 10,000 answers in 2D, figure shows convex hulls of 3 questions



Part I

Hallucination Detection via Attention Head Spectrum

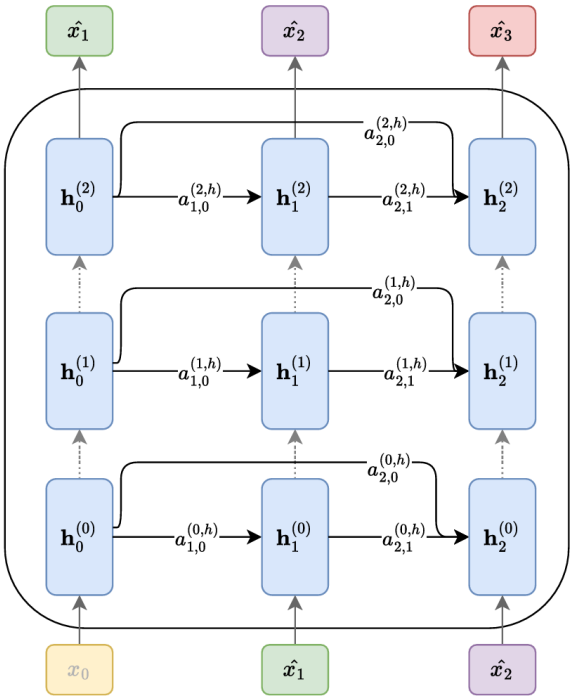
Spectral features from the Transformer's internal attention matrices

Transformers: *Attention is all you need*

Numerical data "flowing" through the decoder

=> Stack of 32 layers and 32 attention heads per layer

This mechanism made modern Generative LLMs possible!



LLM Inference: information flow of 3 generated tokens ("words"). The $a_{ij}^{l,h}$ denotes the attention score of Attention Head (l,h), between the tokens i and j.

Attention Spectrum: Method

1. Extract Spectral Features of Attention Maps

For each layer l and head h, the attention matrix A is viewed as a directed graph. Its graph Laplacian is: $L = D - A$
Eigenvalues lie on the diagonal, bounded in $[-1, 1]$. We retain the top 10 eigenvalues per layer-head block.

2. PCA Compression

With 21 layers × all heads, this yields ~6720 raw spectral features per question. **PCA reduces to 300-500 dimensions**, concentrating useful variance and denoising.

3. Training

Logistic Regression on PCA-projected features achieves the best AUROC. After PCA, the signal is well-captured by a linear boundary: simple probes are effective.

Key finding: combined TriviaQA + MMLU yields the best AUROC. Linear methods outperform tree ensembles.

Extract Spectral features of attentions

1. Extract Attention Maps

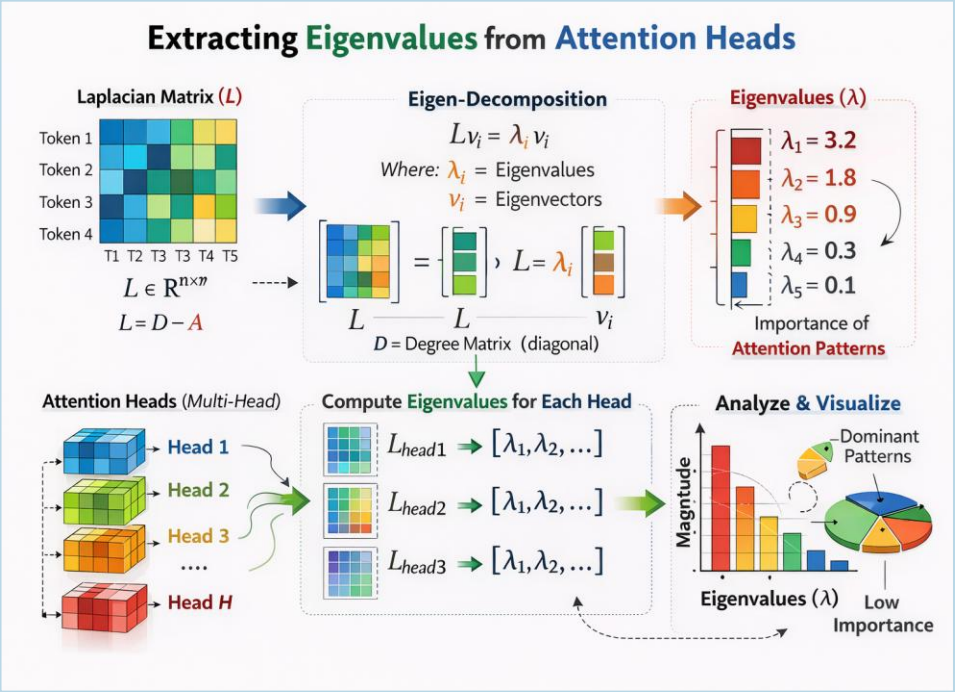
For each layer l and head h, the attention matrix A is viewed as a directed graph. Its graph Laplacian is:

$L = D - A$

Eigenvalues lie on the diagonal, bounded in [−1, 1]. We retain the top 10 eigenvalues per layer-head block.

2. Raw feature story:

- Each layer-head block contributes the top 10 Laplacian eigenvalues.
- Roughly 7000 raw spectral features before compression.
- Raw dimensionality is high and strongly correlated across heads and layers.
- We ask the question: does hallucination leave a measurable signature in model internals?

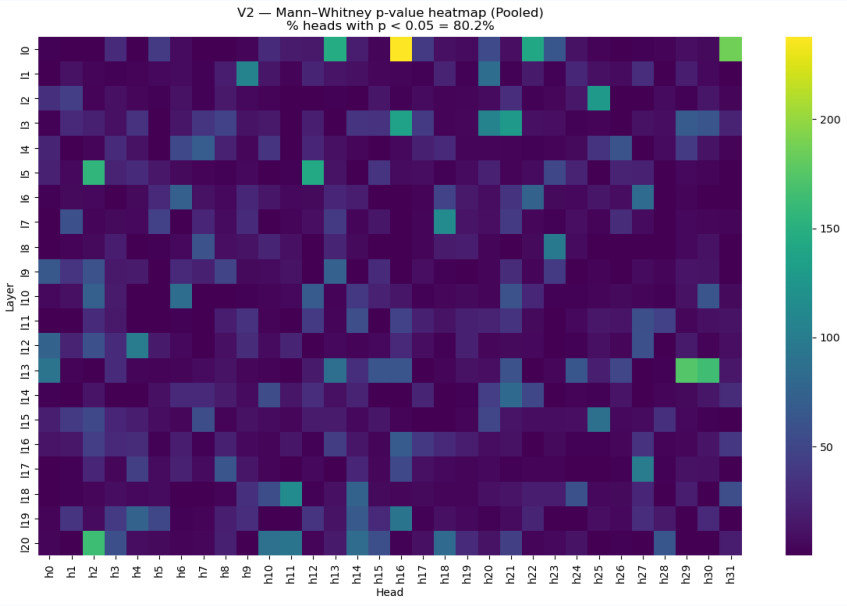


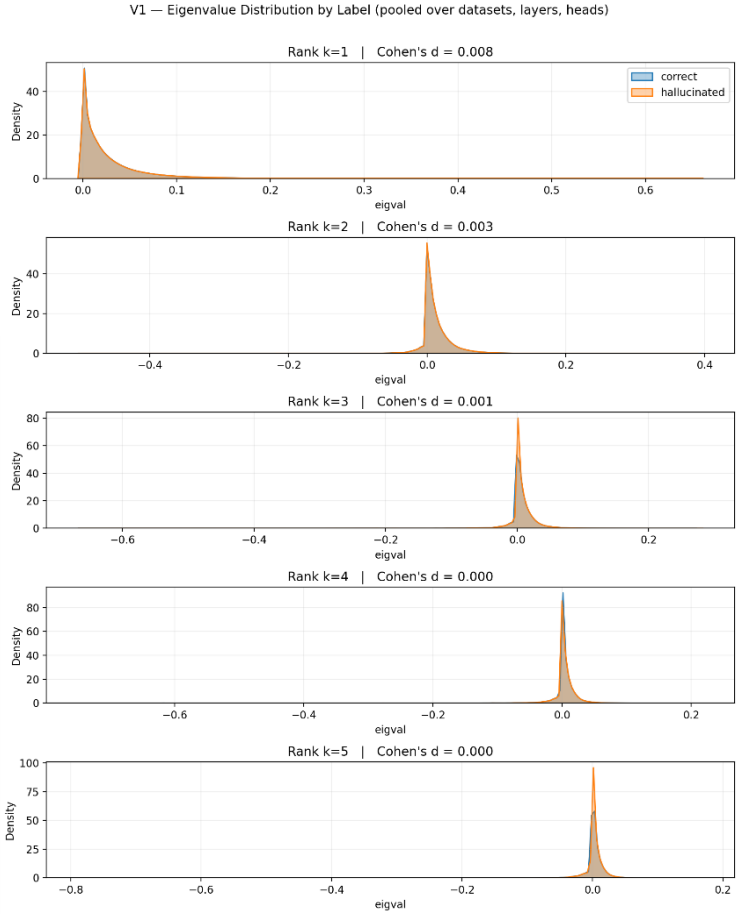
Diagnostics and Statistical Validation

Eigenvalue Distribution by Label & Mann-Whitney p-value Heatmap:

- For each rank (up to 5), plots Laplacian eigenvalue distributions by label: **correct** vs. **hallucinated**.
- In the figure, the strongest separation appears at low ranks, showing that leading eigenvalues already carry label signal.
- Hallucinated samples show broader and heavier-tailed distributions, consistent with more diffuse attention flow.

- The pooled dataset has the highest share of heads with "p<0.05"
- This means spectral differences between correct and hallucinated responses are most consistently detectable when datasets are combined.

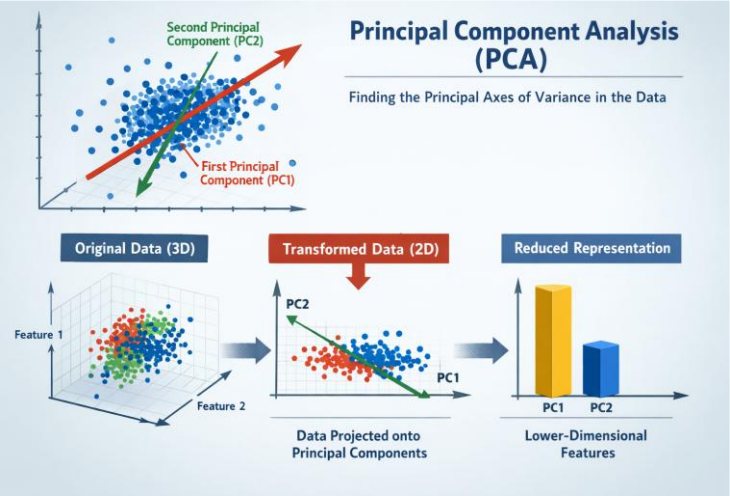


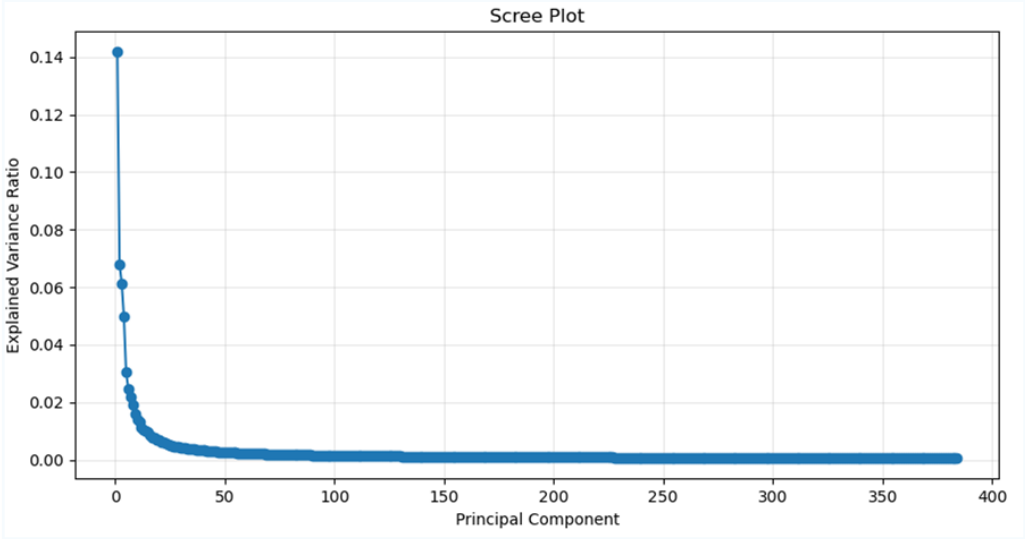


Feature Engineering and PCA

PCA

- Without compression learning is **slower** and **less stable**.
- PCA compresses high-dimensional, noisy, and correlated eigenvalue features into a compact representation.
- Reduce to around **300-500 dimensions** in the main pipeline.
- Act as both **denoising** and **simplification**, not just compression.
- Most of the variance is concentrated in **low dimensions.**





PCA makes the spectral signal more linearly separable. The variance plot suggests hallucination signal is low-dimensional after transformation.

Modeling Results

Training Pipeline: Standardize => PCA => Binary Classifier

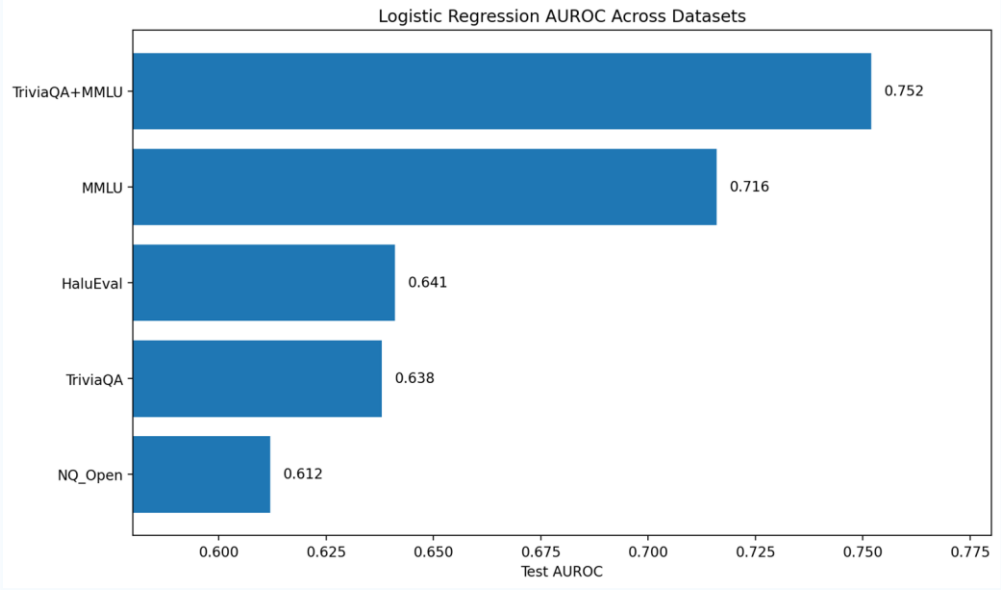
Baseline: Logistic Regression

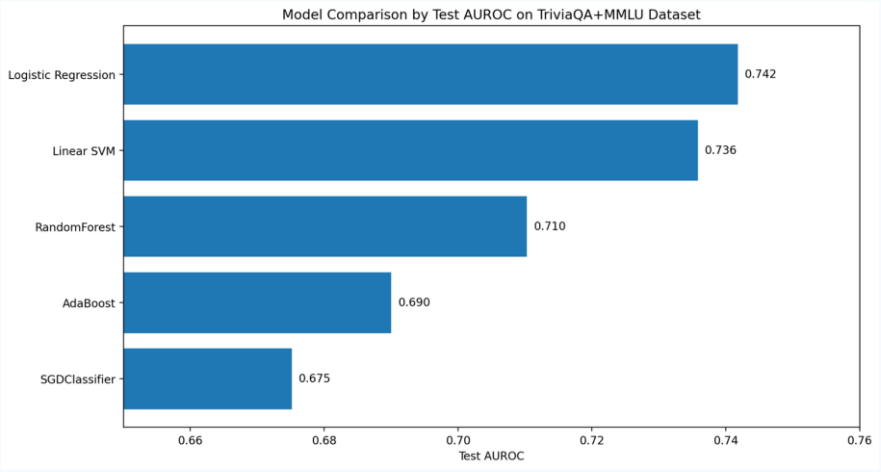
Comparisons: Linear SVM, Random Forest, AdaBoost, SGDClassifier

The best performing dataset is the combined **TriviaQA** and **MMLU** dataset.

The top-performing model is **Logistic Regression**; **Linear SVM is** very close.

Tree ensembles underperform, suggesting that after PCA the signal becomes largely **linearly** separable.





Attention Spectrum: Key Results

0.752

AUROC
Logistic Reg + PCA
(TriviaQA+MMLU)

$p < 0.01$

Bootstrap Test
Hallucinated samples have
higher leading eigenvalue

80.2%

Layer-Head Blocks
with significant signal
($p < 0.05$, Mann-Whitney)

Cross-dataset generalization: training on MMLU+TriviaQA transfers well to NQ-Open. TruthfulQA is an outlier (adversarial phrasing). Deeper layers carry stronger signal, but all-layer model wins overall.

Limitation: Requires access to Transformer internals at runtime. Not available through standard APIs (OpenAI, Anthropic). This motivates Part II.

Part II

Response Cloud Geometry and Semantic Entropy

Black-box hallucination detection from the distribution of sampled responses

Six Unsupervised Geometric Features

Each feature aggregates N=20 response embeddings into a single scalar per question.

Symbol	Name	Formula	Measures
H_sem	Semantic Entropy	$-\sum p_k \ log_2 \ p_k$	How many distinct meanings?
D_cos	Cosine Dispersion	mean(1 − cos(e_j, centroid))	Spread around the center
D_cos_var	Dispersion Variance	var(1 − cos(e_j, centroid))	Asymmetric scatter
D_pair	Pairwise Distance	mean(1 − S_{ij}) over all pairs	Average response separation
K	Cluster Count	# agglom. Clusters	Number of distinct answers
σ^2_S	Similarity Variance	var(S_{ij}) ; over all pairs	Unevenness of agreement

All features are unsupervised: they measure internal consistency of the response cloud without using correctness labels. No reference or known-correct answers are needed.

The Core Intuition

Confident LLM (correct)

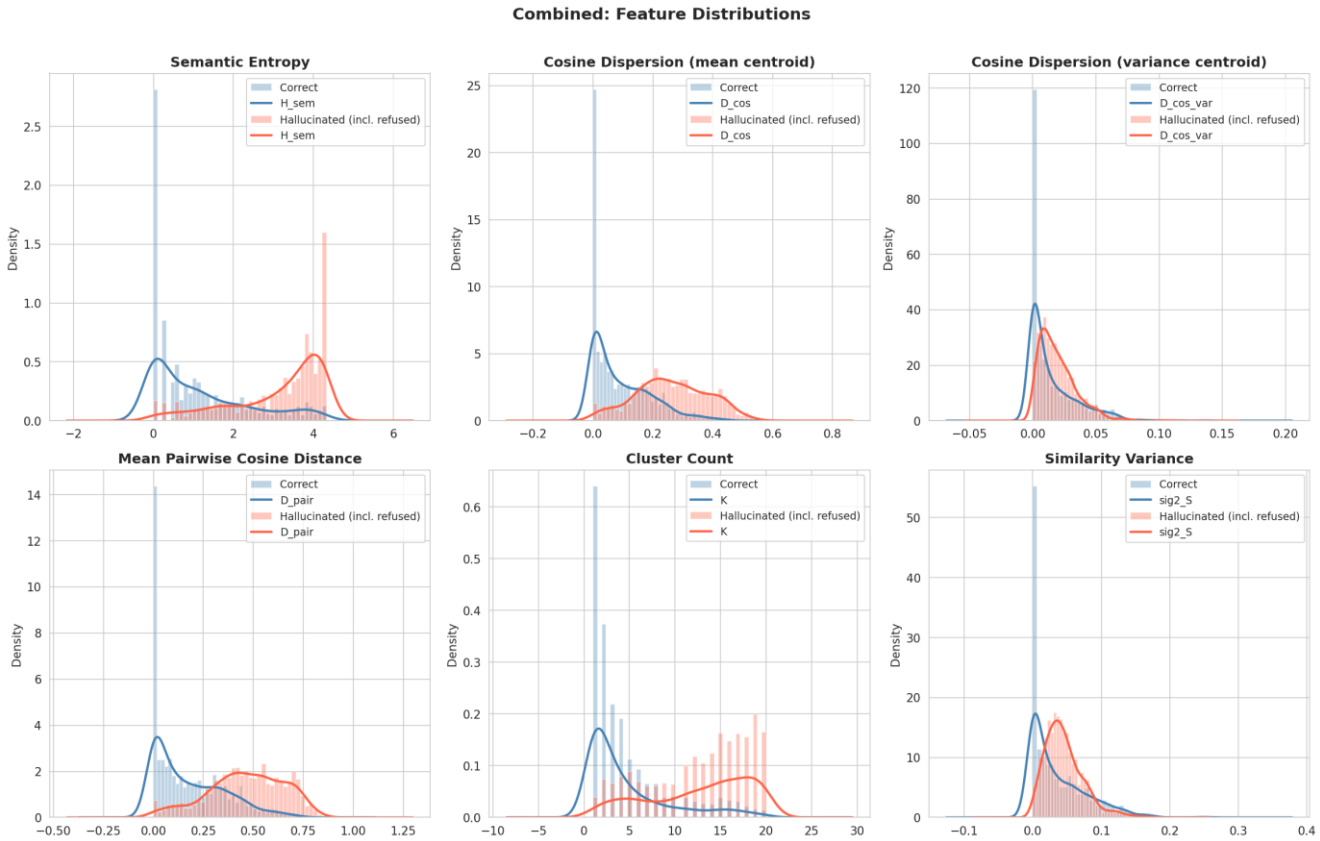
All 20 responses say the same thing
$H_sem \approx 0$ (one cluster)
$D_cos \approx 0.02$ (tight blob)
$K = 1$

Uncertain LLM (hallucinating)

Responses scatter across 5+ meanings
$H_sem \approx 2.0$ (high entropy)
$D_cos \approx 0.25$ (dispersed cloud)
$K = 5$ to 7

Our Hallucination Detector: learn this pattern from 2,500 labeled examples.

Combined Feature Distributions (2500 questions)



Statistical Validation: KS Tests (Combined, 2500 q)

Kolmogorov-Smirnov test per feature: is the distribution different for hallucinated vs correct questions?
Bonferroni correction: $\alpha_adj = 0.05 / 6 = 0.0083$

Feature	KS Statistic	p-value	Significant?
H_sem	0.5927	9.80e-202	Yes ★★★
D_cos	0.5603	5.87e-179	Yes ★★★
D_cos_var	0.3323	8.21e-61	Yes ★★★
D_pair	0.5603	5.87e-179	Yes ★★★
K	0.5792	5.17e-192	Yes ★★★
σ^2_S	0.3382	4.84e-63	Yes ★★★

All 6 features are highly significant after Bonferroni correction. H_sem and K show the strongest separation (D > 0.57), directly validating Farquhar et al. (2024).

(Both tests on the combined 2,500-question dataset)

Permutation Test and Bootstrap AUC

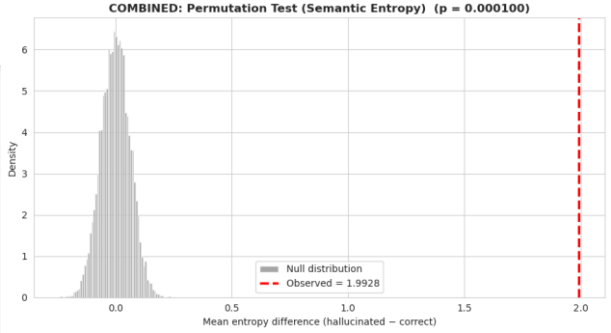
Permutation Test (10,000 shuffles)

H_0: mean entropy is the same for hallucinated and correct questions.

$\Delta = 1.99$ bits

$p < 0.0001$

Hallucinated questions have nearly 2 bits higher entropy. No permutation out of 10,000 matched this gap.



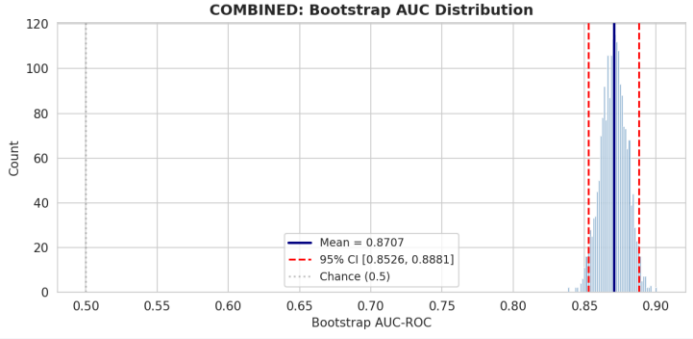
Bootstrap AUC (B=2,000)

Random Forest on 6 features.
Out-of-bag AUC collected per resample.

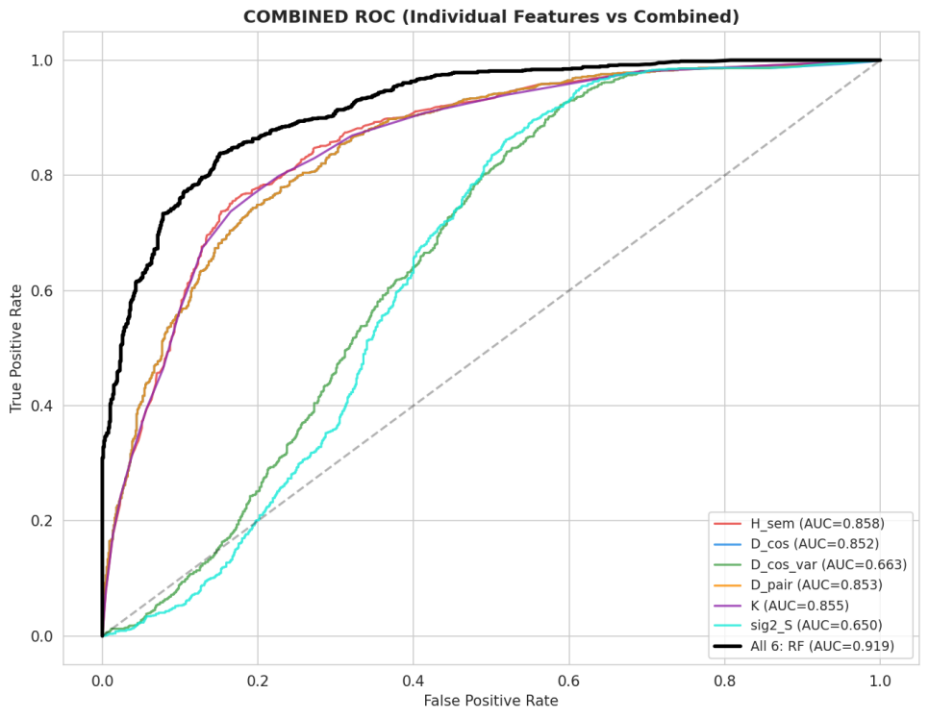
AUC = 0.871

95% CI [0.853, 0.888]

Entire CI above 0.5: the classifier reliably outperforms chance. Narrow CI (width 0.035): stable across resamples.



ROC Curves: Individual Features vs Combined



Black: All 6 features (RF, AUC=0.919)

H_sem, D_cos, D_pair, K each individually achieve AUC > 0.85.

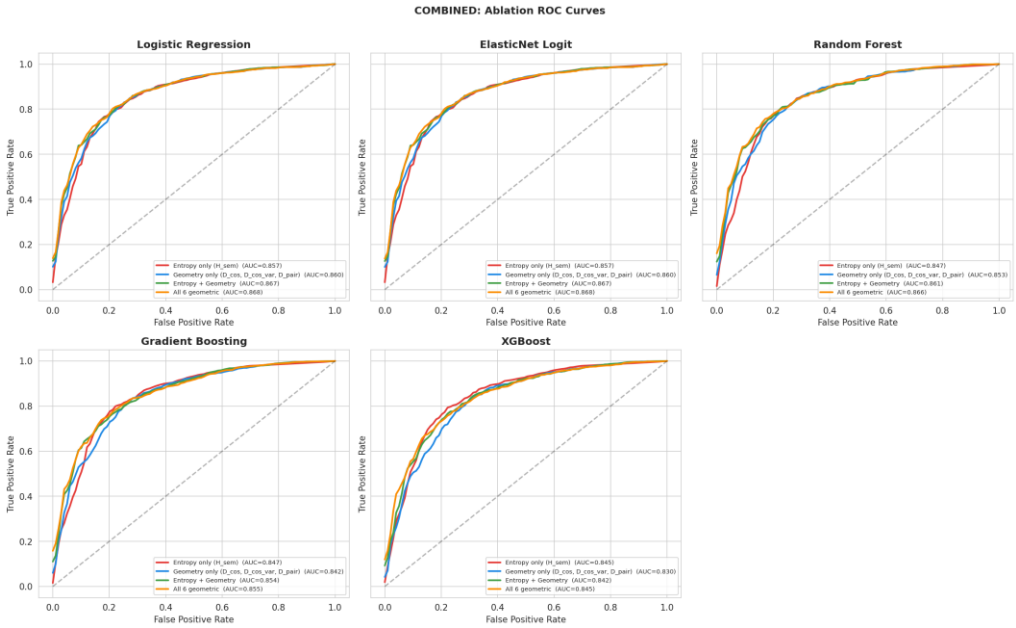
D_cos_var and sig2_S are weaker individually (AUC ~0.65) but add complementary signal.

Combined 6-feature model lifts AUC to 0.92, well above any individual feature.

Ablation Study: Combined (5-fold CV)

Feature Variant	ElasticNet	Gradient Boost	Logistic Reg	Random Forest	XGBoost
All 6 geometric	0.906	0.893	0.906	0.885	0.876
Entropy + Geometry	0.903	0.893	0.904	0.877	0.874
Entropy only (H_sem)	0.861	0.852	0.861	0.843	0.845
Geometry only	0.877	0.868	0.877	0.861	0.859

Best overall: ElasticNet Logit on All 6 features (AUC = 0.906). This aligns with our finding that the hallucination signal is well-captured by logistic-based probes with combined $L_1 + L_2$ penalties. Entropy alone (0.861) already performs well; geometry features add +0.045.



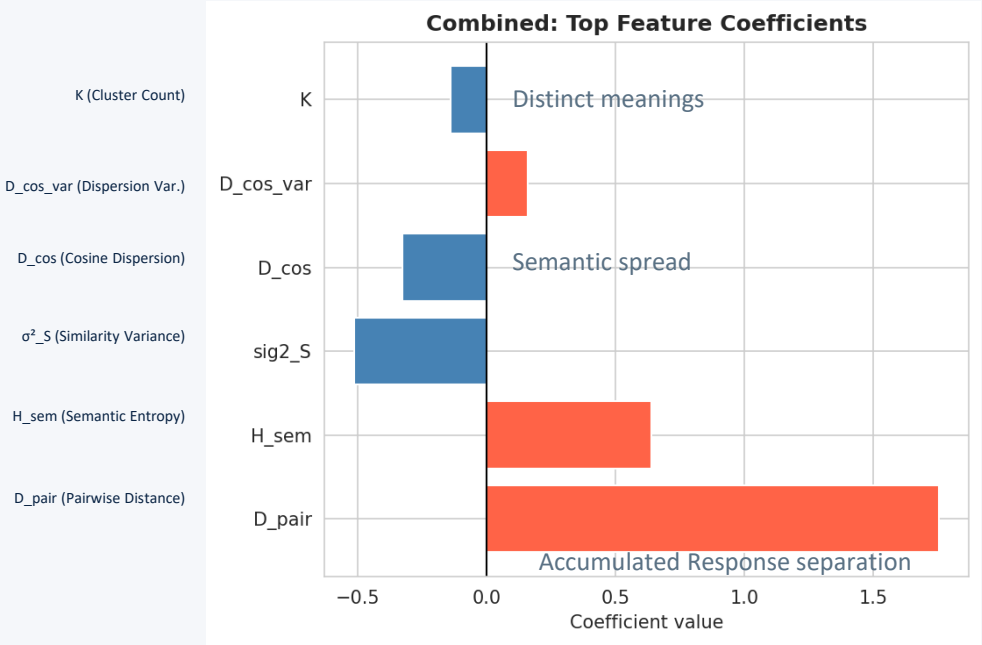
Per-Dataset Results (80/20 Hold-out)

Dataset	Best Variant	Best Classifier	AUC	F1 @ 0.5
DefAn	Geometry only	Logistic Regression	0.823	0.864
HaluEval	All 6	Random Forest	0.948	0.250
MMLU	Geometry only	ElasticNet	0.719	0.862
TriviaQA	Entropy only	Logistic Regression	0.922	0.810
TruthfulQA	Geometry only	Gradient Boosting	0.694	0.824
Combined	**All 6**	**ElasticNet**	**0.906**	**0.843**

Key pattern: Geometry features carry the most signal in harder datasets (DefAn, MMLU, TruthfulQA). Performance improves significantly when training on the combined dataset (0.906 vs 0.69-0.82 individually). The aggregated dataset provides complementary information.

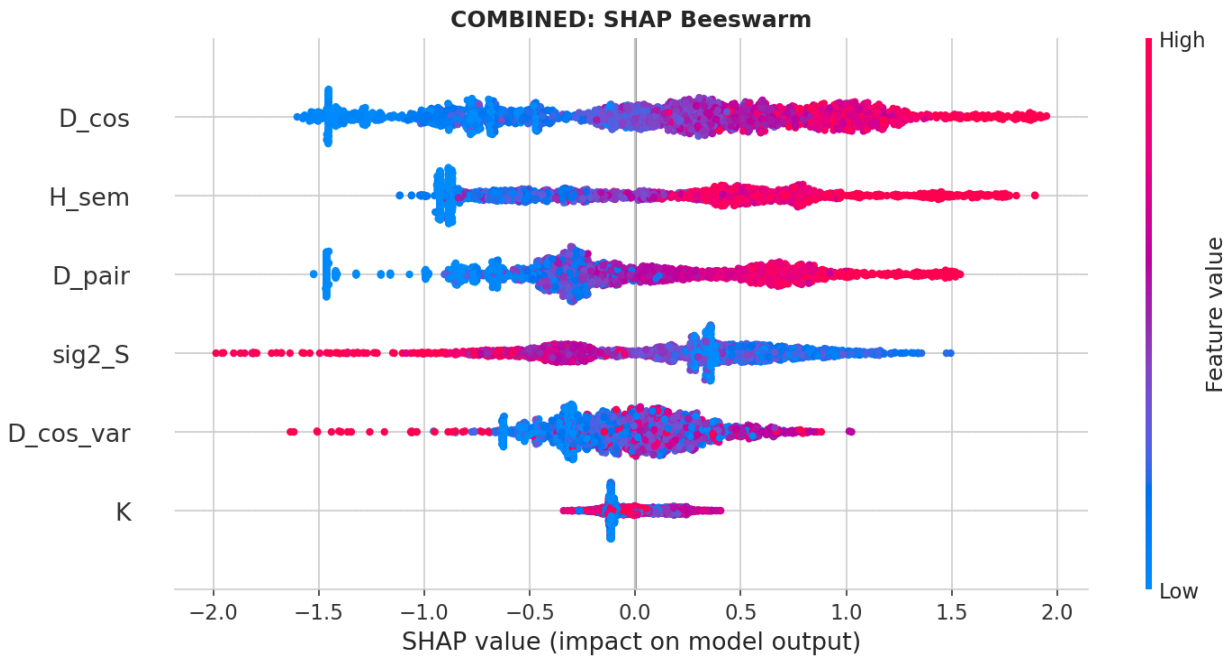
Feature Importance: What Drives Detection?

Top coefficients



Feature Importance: What Drives Detection?

SHAP beeswarm analysis on the combined dataset (2,500 questions)



Red = high feature value pushes prediction toward hallucination.

Blue = low feature value pushes toward correct.

D_cos and H_sem together carry the majority of signal. σ^2_S provides smaller but complementary information.

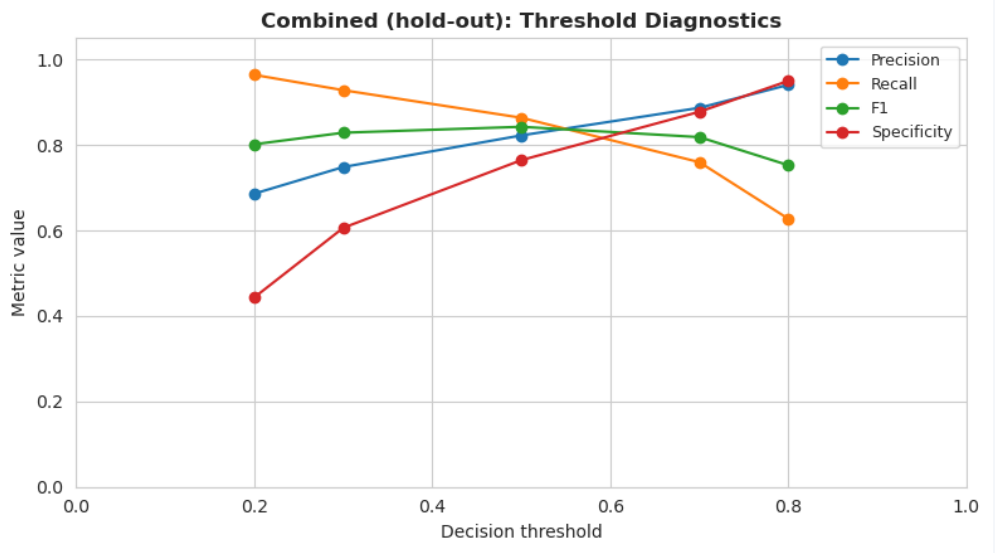
Threshold Diagnostics: Choosing Your Trade-Off

The classifier outputs $p \in [0,1]$. Pick a threshold t to decide: flag if $p \geq t$.

Threshold	Precision	Recall	F1	Use Case
t = 0.2	0.76	0.96	0.85	Safety-critical: catch nearly everything
t = 0.5	0.87	0.84	0.85	General-purpose: balanced trade-off
t = 0.8	0.93	0.59	0.72	High-volume: flag only when very confident

In production: pick t based on the cost ratio of false alarms vs missed hallucinations.

- ✓ Medical/legal: use t=0.2 (never miss a hallucination).
- ✓ Screening: use t=0.8 (only flag what you are sure about).



Some considerations…

A limitation of our pipeline is that we have to repeatedly call on the LLM to generate numerous samples for feature extraction…

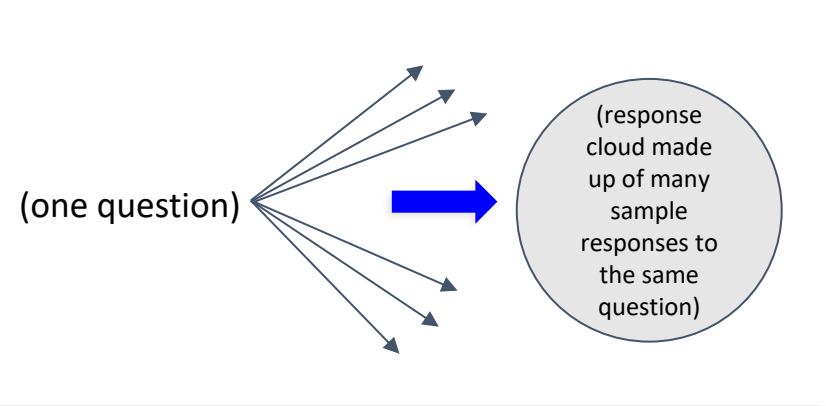
Scale consideration

For real world applications, we would typically want to use **much higher sampling rates** in order to obtain more refined predictions on hallucination risk probability during training. At scale, this can become expensive and / or slow…

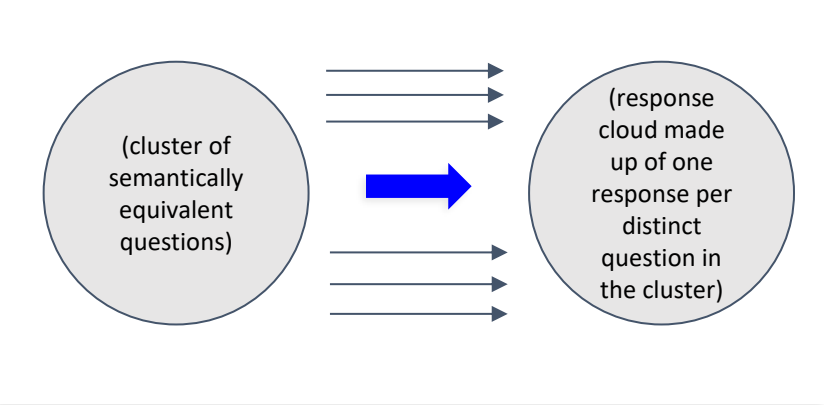
… So, can we use existing data instead?

Companies likely already have swaths of data of the form (user_query, model_answer) from their LLM assistants. So, if we **grouped user queries into clusters by semantic similarity**, we could run our pipeline on the existing data instead of always having to generate fresh samples each run.

At first glance, it seems difficult to test whether we can expect similar results from this, because it is difficult to systematically generate "semantically equivalent questions" from standard QA datasets. Luckily, the DefAn dataset comes pre-loaded with alternative variations for each of its questions, meaning we can use it to simulate this sampling vs. grouping discrepancy locally.



vs.



Feature correlation test

This asks: Do the features computed from sampled response clouds correlate with the same features computed instead from semantic-cluster response clouds? Interpretation: High correlation for a given feature -> that feature is stable whether computed from semantic-clusters or samples. Results: For the main features H_sem, D_cos and sig2, we get high correlations ~0.874, ~0.872, ~0.698 respectively.

Transfer test

This asks: If we learn the mapping f : (features) -> (risk labels) using sample-generated data, does it still work for cluster-generated data? I.e., is $f(X_sample) \approx f(X_cluster)$? Interpretation: If mean CV-AUC from sample -> sample is approximately equal to mean CV-AUC from sample -> cluster, the model learns effectively the same relationship. Results: We obtained mean CV-AUC ~0.801 sample -> sample and mean CV-AUC ~0.799 sample -> cluster. Excellent transference.

Conclusion: Hallucination risk predicted from sampled response data is similar to hallucination risk predicted from semantic cluster data. Thus, our pipeline is effective both ways.

Deploying in Production

1

Sample 20 Responses

Send same prompt 20x at T≥0.7.
Cost: ~$0.001 per question.

2

Embed Locally

all-MiniLM-L6-v2 on CPU.
Milliseconds per response.

3

Compute 6 Features

Pure NumPy. Sub-second.
No external API calls.

4

Classify

ElasticNet outputs $p \in [0,1]$.
Threshold per use case.

Method	Cost per 1M Questions	Latency
GPT-4 Judge (full)	$15,000 - $30,000	~2s per question
GPT-4.1-nano Judge	$2,000	~0.5s per question
Our Detector	**$1,000 + local compute**	**~0.1s classify**

Conclusions

1. Attention-map eigenvalues contain a measurable, transferable signature of hallucination (Part I). PCA + linear probes achieve strong AUROC.

2. Six unsupervised geometric features from the response cloud reliably detect hallucinations as a black-box (Part II). AUC = 0.91 on combined dataset.

3. D_cos (semantic spread) and H_sem (fragmentation) carry the majority of the signal, validating and extending Farquhar et al. (2024).

4. ElasticNet Logit is the best overall classifier: simple logistic probes with $L_1 + L_2$ penalties capture the hallucination signal cleanly.

5. The detector costs ~$0.001 per question and can replace or pre-screen expensive LLM judges, enabling scalable deployment in production AI systems.

Thank you! Adedoapo Ojoawo, Steven Gubkin, Alec Clott, and the Erdos Institute
github.com/TheErdosInstitute/spring-2026-LLM-hallucinations